Applied Econometrics: Session 9

Regression Discontinuity Design

PhD Intensive Course

Lecture 9

April 29, 2026

How This Lecture Fits the Course

Earlier lectures used controls, instruments, matching, DiD, fixed effects, and synthetic controls to construct credible counterfactuals.

Regression discontinuity uses a different source of credibility:

Core idea

If treatment changes discontinuously at a cutoff, units just below and just above the cutoff can be compared locally.

- The running variable is observed and determines eligibility.
- The cutoff creates a jump in treatment probability.
- The causal estimate is local: it applies at the cutoff, not automatically far away.

Roadmap for Today

1. Cutoff Intuition
2. Sharp RD Identification
3. Local Linear Estimation
4. Diagnostics
5. Fuzzy RD
6. Simulation and Review

A Threshold as a Natural Experiment

Examples of cutoff rules:

- students pass an exam if score ≥ 50;
- firms receive a subsidy if employment is below 250 workers;
- people become legally eligible for a program at a certain age;
- municipalities receive extra funding if population is below a threshold.

Why this may help

A person who scores 49.9 and a person who scores 50.1 are probably very similar, except that the rule treats them differently.

The Running Variable

Let R_i be the running variable and c the cutoff.

$$X_i \equiv R_i - c$$

is the centered running variable.

- $X_i < 0$: unit is below the cutoff.
- $X_i = 0$: unit is exactly at the cutoff.
- $X_i > 0$: unit is above the cutoff.

Practical habit

Center the running variable before estimating RD. Then the intercepts of local regressions are values at the cutoff.

Sharp RD Treatment Rule

A sharp RD has deterministic assignment:

$$D_i = 1\{R_i \geq c\} = 1\{X_i \geq 0\}.$$

- Everyone below the cutoff is untreated.
- Everyone above the cutoff is treated.
- The treatment probability jumps from 0 to 1 at c.

Important

Sharp means the rule fully determines treatment. It does not mean the design is automatically valid.

Potential Outcomes Near a Cutoff

Each unit has two potential outcomes:

$$Y_i(1), \qquad Y_i(0).$$

We observe

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0).$$

- Above the cutoff, sharp RD reveals treated outcomes.
- Below the cutoff, sharp RD reveals untreated outcomes.
- The missing object is the counterfactual outcome on the other side of the cutoff.

The Visual Logic

Imagine plotting average outcome against $X_i = R_i - c$.

- Away from the cutoff, treated and untreated units may be very different.
- Very close to the cutoff, the running variable is nearly the same.
- A discontinuous jump in the outcome at $X = 0$ is evidence of a treatment effect.

RD picture in words

Fit the smooth left-hand relationship up to the cutoff and the smooth right-hand relationship down to the cutoff. The vertical gap at zero is the RD estimate.

What RD Identifies

RD is not mainly about comparing all treated units with all untreated units.

$$\text{all above cutoff} \quad \text{vs.} \quad \text{all below cutoff}$$

is usually a bad comparison because high-R and low-R units differ.

RD is about the limit comparison:

$$R_i \downarrow c \quad \text{vs.} \quad R_i \uparrow c.$$

Local target

RD estimates the treatment effect for units at the cutoff, or for units close enough that the cutoff comparison is credible.

Why “Barely Passed” Is Not “Passed by a Lot”

Suppose an exam score determines graduation.

- Score 90 vs. score 40: very different students.
- Score 50.1 vs. score 49.9: likely similar students.
- The cutoff comparison removes much of the concern that ability drives the result.

Student warning

RD is credible because it is narrow. Do not turn a local design into a broad claim without extra evidence.

The Continuity Assumption

For $d \in \{0, 1\}$, define

$$\mu_d(r) = \mathbb{E}[Y_i(d) \mid R_i = r].$$

The key identifying assumption is:

$$\lim_{r \to c^-} \mu_d(r) = \lim_{r \to c^+} \mu_d(r), \qquad d = 0, 1.$$

- Potential outcomes may slope with the running variable.
- They must not jump at the cutoff for reasons other than treatment.
- The observed outcome can jump because treatment jumps.

Appendix details

Sharp RD Estimand

Under continuity and sharp assignment:

$$\tau_{\mathrm{SRD}} = \lim_{r \to c^+} \mathbb{E}[Y_i \mid R_i = r] - \lim_{r \to c^-} \mathbb{E}[Y_i \mid R_i = r].$$

This equals

$$\mathbb{E}[Y_i(1) - Y_i(0) \mid R_i = c].$$

Interpretation

The estimand is a treatment effect at the cutoff, not necessarily the average treatment effect for the full population.

Appendix details

Local Randomization Intuition

Near the cutoff, small differences in R_i can be as good as random.

- A student may barely pass or barely fail because of small test-day noise.
- A person may be just above or below an age threshold because of date of birth.
- A firm may just exceed an administrative size threshold.

But be precise

Local randomization is an intuition. The formal RD assumption is continuity of potential outcomes at the cutoff.

What Can Break Continuity?

Continuity fails if something else jumps at the same cutoff.

- Another policy starts at exactly the same threshold.
- Measurement of the outcome changes at the cutoff.
- People sort around the threshold using private information.
- The cutoff itself is chosen because expected outcomes jump there.

Design question

If treatment did not change at the cutoff, would the expected outcome have moved smoothly through it?

A Numeric Cutoff Example

Imagine a scholarship for scores ≥ 70.

	Mean outcome just below 70	Mean outcome just above 70
Observed graduation rate	0.62	0.69

The sharp RD estimate is

$$0.69 - 0.62 = 0.07.$$

Correct wording

The scholarship increased graduation by 7 percentage points for students at the score cutoff, if potential graduation rates would otherwise have been continuous at 70.

What Not to Estimate

A global mean comparison:

$$\mathbb{E}[Y_i \mid R_i \geq c] - \mathbb{E}[Y_i \mid R_i < c]$$

is generally not an RD estimand.

- Units far above and far below the cutoff can be very different.
- The running variable itself often strongly predicts the outcome.
- RD asks for a local discontinuity, not a global level difference.

Exam phrase

"Close to the cutoff" is not a cosmetic detail. It is the source of identification.

Center First

Define

$$X_i = R_i - c, \qquad Z_i = 1\{X_i \geq 0\}.$$

Centering makes the cutoff equal to zero.

- Left of cutoff: $Z_i = 0$.
- Right of cutoff: $Z_i = 1$.
- At $X_i = 0$, slopes vanish and intercepts have a direct cutoff interpretation.

Local Linear RD Regression

A standard sharp RD specification is

$$Y_i = \beta_0 + \beta_1 X_i + \tau Z_i + \beta_2 (Z_i X_i) + u_i.$$

- Below cutoff: $\mathbb{E}[Y_i \mid X_i] = \beta_0 + \beta_1 X_i$.
- Above cutoff: $\mathbb{E}[Y_i \mid X_i] = (\beta_0 + \tau) + (\beta_1 + \beta_2) X_i$.
- At $X_i = 0$, the right-left jump is τ.

Interpretation

This is like fitting two lines, one on each side, and comparing their intercepts at the cutoff.

Appendix details

Why Local, Not Global?

If the true relationship between Y and R is curved, a line fit far from the cutoff can distort the predicted value at the cutoff.

- A very wide window gives more observations but more functional-form bias.
- A very narrow window gives less bias but noisier estimates.
- RD estimation is therefore a bias-variance tradeoff.

Practical rule

The chosen bandwidth should be reported, justified, and varied in robustness checks.

Bandwidth

Use observations with

$$|X_i| \leq h.$$

The bandwidth h controls how local the comparison is.

- Small h: close comparison, high variance.
- Large h: more precision, more extrapolation.
- Main estimates should not rely on one fragile bandwidth choice.

Student habit

Always ask: how many observations are inside the bandwidth on each side?

Triangular Kernel Weights

A common local weighting rule is the triangular kernel:

$$K_i(h) = 1\{|X_i| \leq h\} \left(1 - \frac{|X_i|}{h}\right).$$

- Units outside the bandwidth get zero weight.
- Units near the cutoff get the largest weight.
- Units near the edge of the window get small weight.

Why this matches RD logic

The closest observations are the most credible comparisons.

Weighted Local Linear Estimator

Estimate

$$\min_{\beta_0,\beta_1,\tau,\beta_2} \sum_i K_i(h)\left[Y_i - \beta_0 - \beta_1 X_i - \tau Z_i - \beta_2 Z_i X_i\right]^2 .$$

The coefficient on Z_i is the estimated jump at the cutoff.

Key point

The regression is only an interpolation device near the cutoff. The design, not the linear model, is doing the causal work.

Appendix details

Why Local Linear Is the Workhorse

Local linear RD is popular because:

- it allows different slopes on the two sides;
- it focuses on the boundary point $X = 0$;
- it has better boundary behavior than a simple local mean;
- it is easy to explain and graph.

Caution

High-order global polynomials can create artificial wiggles. They are usually a poor substitute for local estimation.

Standard Errors

RD uncertainty comes from estimating two limits with finite data.

- Use heteroskedasticity-robust standard errors as a default.
- If data are clustered, cluster at the assignment or sampling level.
- If the running variable is discrete with few support points, inference needs extra care.

Reporting

Report the point estimate, standard error, confidence interval, bandwidth, kernel, and observations on both sides.

Graphical RD Plot

A useful RD graph has:

- binned means of Y_i against centered X_i;
- a vertical line at the cutoff;
- separate fitted curves or local lines on each side;
- enough scale information to see the amount of data near zero.

Purpose

The graph should show whether the estimated jump is visible and whether the functional form looks plausible near the cutoff.

Covariate Balance at the Cutoff

Pre-treatment covariates should not jump at the cutoff.

$$\lim_{r \to c^+} \mathbb{E}[W_i \mid R_i = r] - \lim_{r \to c^-} \mathbb{E}[W_i \mid R_i = r] \approx 0$$

for predetermined W_i.

- Age, prior test scores, pre-policy outcomes, baseline demographics.
- A covariate jump suggests sorting, another policy, or measurement problems.

Do not overstate

Balance tests cannot prove validity. They can reveal obvious failures.

Manipulation and Density Checks

The density of the running variable should be smooth through the cutoff.

- If people can precisely manipulate R_i, many may bunch just on the favorable side.
- Examples: reported income thresholds, exam retakes, administrative score rounding.
- A density jump weakens the "barely below vs. barely above" comparison.

McCrary-style idea

Check whether the number of observations changes discontinuously at the cutoff.

Appendix details

Placebo Checks

RD placebo checks ask whether jumps appear where they should not.

- Use fake cutoffs where treatment does not change.
- Use pre-treatment outcomes as placebo outcomes.
- Use predetermined covariates as placebo outcomes.

Interpretation

A placebo failure does not tell you exactly what is wrong, but it tells you the cutoff is not behaving like a clean local experiment.

Donut RD

Sometimes observations extremely close to the cutoff are suspicious.

- Scores may be rounded or manually adjusted.
- People may manipulate just enough to cross the rule.
- Administrative discretion may concentrate at the threshold.

A donut RD drops a small interval around the cutoff:

$$\delta < |X_i| \leq h.$$

Use

Donut RD is a sensitivity check, not a license to ignore manipulation.

RD Reporting Checklist

A credible RD write-up should state:

- running variable and cutoff;
- why the rule changes treatment at that cutoff;
- sharp or fuzzy design;
- local estimand and population near the cutoff;
- bandwidth, kernel, and polynomial order;
- RD plot, covariate balance, and density/manipulation evidence;
- robustness to bandwidths and reasonable specifications.

When RD Is Fuzzy

A fuzzy RD occurs when treatment probability jumps at the cutoff but not from 0 to 1.

$$\lim_{r \to c^+} \mathbb{P}(D_i = 1 \mid R_i = r) \neq \lim_{r \to c^-} \mathbb{P}(D_i = 1 \mid R_i = r)$$

but both probabilities may be between 0 and 1.

- Some eligible people do not take treatment.
- Some ineligible people receive treatment anyway.
- The cutoff is an instrument for treatment receipt.

Examples of Fuzzy RD

- Legal drinking age increases access to alcohol, but some underage people drink.
- Passing an exam increases diploma receipt, but some failing students later receive credentials.
- Program eligibility increases participation, but some eligible people do not enroll.
- A scholarship cutoff increases aid receipt, but take-up is incomplete.

Translation

Sharp RD estimates the effect of eligibility if eligibility equals treatment. Fuzzy RD estimates the effect of treatment for cutoff compliers.

Cutoff as an Instrument

Define the cutoff indicator:

$$Z_i = 1\{R_i \geq c\}.$$

In fuzzy RD, Z_i is an instrument for actual treatment D_i.

- First stage: Z_i changes treatment probability at c.
- Reduced form: Z_i changes the outcome at c.
- Wald ratio: reduced-form jump divided by first-stage jump.

Connection to Lecture 5

Fuzzy RD is IV where the instrument is local cutoff eligibility.

Fuzzy RD Estimand

The fuzzy RD estimand is

$$\tau_{\mathrm{FRD}} = \frac{\lim_{r \to c^+} \mathbb{E}[Y_i \mid R_i = r] - \lim_{r \to c^-} \mathbb{E}[Y_i \mid R_i = r]}{\lim_{r \to c^+} \mathbb{E}[D_i \mid R_i = r] - \lim_{r \to c^-} \mathbb{E}[D_i \mid R_i = r]}.$$

Under the usual IV assumptions locally at the cutoff, this is

$$\mathbb{E}[Y_i(1) - Y_i(0) \mid \text{complier at } c].$$

Appendix details

Fuzzy RD Assumptions

For a local Wald interpretation, we need:

1. continuity of potential outcomes at the cutoff;
2. a nonzero first-stage jump in treatment;
3. exclusion: crossing the cutoff affects Y only through D;
4. monotonicity: crossing the cutoff does not make anyone less likely to receive treatment.

Interpretation

The estimate is local twice: local to the cutoff and local to compliers whose treatment status is changed by crossing the cutoff.

Numeric Fuzzy RD Example

	Just below cutoff	Just above cutoff
$\mathbb{E}[Y \mid R]$	50	56
$\mathbb{E}[D \mid R]$	0.30	0.80

Reduced-form jump:

$$56 - 50 = 6.$$

First-stage jump:

$$0.80 - 0.30 = 0.50.$$

Fuzzy RD estimate:

$$\frac{6}{0.50} = 12.$$

Sharp vs. Fuzzy RD

Feature	Sharp RD	Fuzzy RD
Treatment rule	$D_i = 1\{R_i \geq c\}$	cutoff shifts probability of D_i
Jump used	outcome jump	outcome jump divided by treatment jump
Interpretation	effect at cutoff	complier effect at cutoff
Analogy	local experiment	local IV design
Main danger	discontinuity from other causes	weak first stage or exclusion failure

Python Example for This Lecture

The companion script is:

`lectures/code/lecture-9-rdd.py`.

It simulates:

- sharp RD data with a known treatment effect at the cutoff;
- local linear estimates under several bandwidths;
- fuzzy RD with incomplete take-up and a local Wald estimate;
- covariate and density diagnostics for manipulation checks.

Use in class

Ask students to change the bandwidth and curvature, then predict how bias and variance should move before running the script.

What the Simulation Should Teach

- The global treated-control gap is not the RD estimate.
- Local linear estimates get closer to the true cutoff effect when the window is reasonable.
- Very narrow windows can be unstable because little data remain.
- In fuzzy RD, the outcome jump must be scaled by the treatment jump.
- Manipulation shows up as suspicious bunching in the running variable.

Common Mistakes

- Calling any threshold regression an RD without showing treatment jumps at the cutoff.
- Comparing all observations above and below the cutoff.
- Forgetting to center the running variable before interpreting the intercept.
- Using high-order global polynomials as the main design.
- Ignoring density and covariate balance diagnostics.
- Treating fuzzy RD as sharp RD.
- Claiming an ATE for everyone when the design identifies a cutoff effect.

Main Takeaways

1. RD uses a cutoff rule to construct a local comparison.
2. Sharp RD identifies the outcome jump at the cutoff as the treatment effect under continuity.
3. Local linear regression estimates the two one-sided limits.
4. Bandwidth choice controls the bias-variance tradeoff.
5. Diagnostics should check graphs, covariates, and density around the cutoff.
6. Fuzzy RD is a local IV design: reduced form divided by first stage.

Source and Further Reading

This lecture is based on ideas from:

- Matheus Facure Alves, *Causal Inference for the Brave and True*, Chapter 16, "Regression Discontinuity Design".
- Angrist and Pischke, *Mostly Harmless Econometrics*, chapter on regression discontinuity.
- Angrist and Pischke, *Mastering 'Metrics*, chapter on regression discontinuity.
- Lee and Lemieux (2010), "Regression Discontinuity Designs in Economics".

Handbook URL:

- `https://matheusfacure.github.io/python-causality-handbook/16-Regression-Discontinuity-Design.html`

What Students Should Be Able to Do

After this lecture, students should be able to:

- identify the running variable, cutoff, and treatment rule;
- write the sharp RD estimand using one-sided limits;
- explain the continuity assumption in words;
- estimate and interpret a local linear RD coefficient;
- explain the role of bandwidth and kernel weights;
- distinguish sharp and fuzzy RD;
- describe graphical, covariate, and manipulation diagnostics.

Appendix Map

- A1: Sharp RD identification.
- A2: Local linear RD coefficient.
- A3: Bandwidth bias-variance logic.
- A4: Fuzzy RD as local Wald/IV.
- A5: Density manipulation check.

Appendix A1: Sharp RD Identification

In a sharp RD,

$$D_i = 1\{R_i \geq c\}.$$

Therefore,

$$\lim_{r \to c^+} \mathbb{E}[Y_i \mid R_i = r] = \lim_{r \to c^+} \mathbb{E}[Y_i(1) \mid R_i = r],$$

and

$$\lim_{r \to c^-} \mathbb{E}[Y_i \mid R_i = r] = \lim_{r \to c^-} \mathbb{E}[Y_i(0) \mid R_i = r].$$

By continuity of $\mathbb{E}[Y_i(d) \mid R_i = r]$ at c,

$$\lim_{r \to c^+} \mathbb{E}[Y_i(1) \mid R_i = r] = \mathbb{E}[Y_i(1) \mid R_i = c],$$

$$\lim_{r \to c^-} \mathbb{E}[Y_i(0) \mid R_i = r] = \mathbb{E}[Y_i(0) \mid R_i = c].$$

Subtracting gives

$$\tau_{\mathrm{SRD}} = \mathbb{E}[Y_i(1) - Y_i(0) \mid R_i = c].$$

Back to main slide

Appendix A2: Local Linear RD Coefficient

With $X_i = R_i - c$ and $Z_i = 1\{X_i \geq 0\}$, estimate

$$Y_i = \beta_0 + \beta_1 X_i + \tau Z_i + \beta_2 Z_i X_i + u_i.$$

For $X_i < 0$, $Z_i = 0$, so

$$\mathbb{E}[Y_i \mid X_i = x, Z_i = 0] = \beta_0 + \beta_1 x.$$

Taking the left limit at zero gives

$$\lim_{x \to 0^-} \mathbb{E}[Y_i \mid X_i = x] = \beta_0.$$

For $X_i \geq 0$, $Z_i = 1$, so

$$\mathbb{E}[Y_i \mid X_i = x, Z_i = 1] = (\beta_0 + \tau) + (\beta_1 + \beta_2)x.$$

Taking the right limit at zero gives $\beta_0 + \tau$. The jump is τ.

Back to main slide

Appendix A3: Bandwidth Bias-Variance Logic

Let $\widehat{\tau}(h)$ be the RD estimate using bandwidth h. A useful mental decomposition is

$$\widehat{\tau}(h) - \tau = \underbrace{\text{bias}(h)}_{\text{curvature/extrapolation}} + \underbrace{\text{noise}(h)}_{\text{sampling variation}} .$$

- As h increases, more observations enter, so variance usually falls.
- As h increases, observations farther from the cutoff influence the fit, so bias may rise.
- As h shrinks, bias usually falls, but the estimate becomes noisy.

$$\text{MSE}(h) = \text{bias}(h)^2 + \mathrm{Var}(\widehat{\tau}(h)).$$

Back to main slide

Appendix A4: Fuzzy RD as Local Wald

Let $Z_i = 1\{R_i \geq c\}$. Define the local reduced-form jump:

$$\Delta_Y = \lim_{r \to c^+} \mathbb{E}[Y_i \mid R_i = r] - \lim_{r \to c^-} \mathbb{E}[Y_i \mid R_i = r].$$

Define the local first-stage jump:

$$\Delta_D = \lim_{r \to c^+} \mathbb{E}[D_i \mid R_i = r] - \lim_{r \to c^-} \mathbb{E}[D_i \mid R_i = r].$$

Under continuity, exclusion, relevance, and monotonicity,

$$\Delta_Y = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i(1) > D_i(0), R_i = c]\Delta_D.$$

Hence

$$\frac{\Delta_Y}{\Delta_D} = \mathbb{E}[Y_i(1) - Y_i(0) \mid \text{complier at } c].$$

Back to main slide

Appendix A5: Density Manipulation Check

Let $f_R(r)$ be the density of the running variable. A manipulation check asks whether

$$\lim_{r \to c^-} f_R(r) = \lim_{r \to c^+} f_R(r).$$

If the density jumps at c, units may be sorting around the threshold.

A simple classroom approximation:

1. bin the running variable into narrow intervals;
2. count observations in each bin;
3. look for a discontinuous count jump at the cutoff.

Formal McCrary-style tests use local smoothing of the log density on both sides.

Back to main slide